

CANONICAL LESSON

Core Agricultural Indicators With Denominators

Decision-centred foundations for accountable sector intelligence

Production Unit	PU-B07-C09
Notebook	V7_A_N04
Status	Controlled educational draft v0.1
Data boundary	Synthetic instructional data only

Central decision: Publish agricultural indicators only with explicit numerators, denominators, populations, reference periods, geographic units, and revision status.

Learning outcomes

Why this matters

Indicators without defensible denominators invite false comparisons across crops, districts, seasons, and reporting systems.

The analytical workflow is useful only when evidence, uncertainty, authority, and action are connected explicitly. The model supports professional judgement; it does not replace accountable institutional authority.

Worked cross-sector example

A crop-loss rate identifies affected hectares as the numerator and assessed planted hectares—not all land—as the denominator.

Boundary: The example supports learning and review. It does not constitute an official statistic, eligibility determination, enforcement action, clinical judgement, or operational authorization.

Practice and reflection

- 1.
- Distinguish mean and ratio yield.
- 2.

- Add confidence intervals.
- 3.
- Explain why indicators cannot set budgets.

Answer key

- 1.
- Mean yield averages holding yields; ratio yield divides estimated total production by total area and answers a different question.
- 2.
- Use the survey design, clustering, stratification, and weights.
- 3.
- Budgets also require objectives, costs, equity, constraints, alternatives, authority, and uncertainty.

Operational readiness checklist

- Purpose and decision owner are named.
- Evidence definitions, provenance, and lawful access are documented.
- Uncertainty and abstention are visible.
- Human review, escalation, monitoring, incident response, and retirement are assigned.
- Equity, accessibility, security, and sector-specific approval are complete.